Module IV

Descriptive Statistics an overview

It refers to the methods and techniques used to summarize and describe the main features of a data set. It involves organizing, displaying and analyzing data to provide simple summaries.

Main components of descriptive statistics

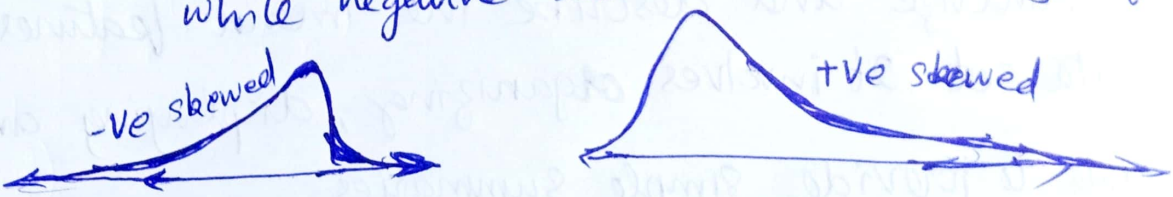
1. Measures of central tendency
   - 1.1 Mean: The average of all datapoints in a data set
   - 1.2 Median: The middle value when the data is ordered from smallest to largest
   - 1.3 Mode: The most frequent values in a data set
2. Measures of dispersion (variability)
   - 2.1 Range: The difference between the largest and smallest values in the data set.
   - 2.2 Variance: A measure of how much the data points deviate from the mean, squared.
   - 2.3 Standard deviation: The square root of the variance representing the spread of the data points around the mean.
   - 2.4. $IQR = Q_3 - Q_1$. Inter quartile range

     The difference between 1st Quartile Q_1 and 3rd quartile Q_3, representing the middle 50% of data.

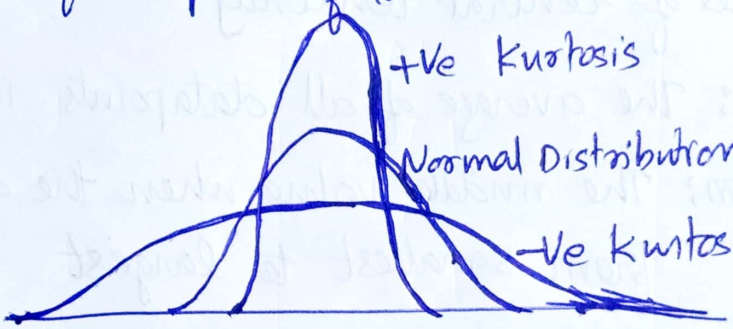
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3. Measures of shape

3.1. Skewness: Describes the asymmetry of the data distribution. Positive indicates a long right tail while negative skew indicates a long left tail.

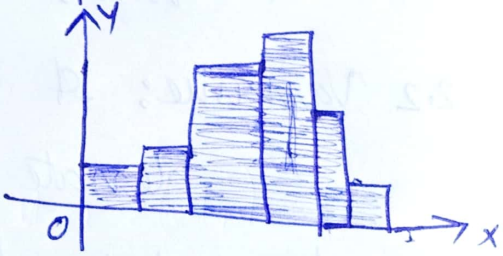


3.2. Kurtosis: Measures the tailedness or sharpness of the peaks of the data distribution.



4. Frequency Distribution.

4.1. Histogram: A bar chart that displays the frequency of data points within specified ranges.



4.2 Frequency table:

A table showing the number of occurences of each distinct value in the data set.

Fruit	Tally Marks	Frequency
Orange	𝍸	5
Apple	𝍸 |	6
Mango	|||	3
		TOTAL = 14

5. Percentile and Quartiles

5.1. Percentiles:- Values that divide the data into 100 equal parts

5.2 Quartiles: Special percentiles that divide the data into 4 equal parts including the first quartile Q_1, second quartile, median Q_2 and third quartile Q_3.

6. Measures of Association - correlation.

correlation refers to a statistical relationship between two variables, indicating the extent to which they tend to vary together. A positive correlation means that as one variable increases, the other tends to increase as well, a negative correlation means that as one variable increases, the other tends to decrease



6.1 Karl Pearson's coefficient of correlation, R:

$$r = \frac{n\Sigma xy - [\Sigma x][\Sigma y]}{\sqrt{n\Sigma x^2 - (\Sigma x)^2}\sqrt{n\Sigma y^2 - (\Sigma y)^2}} \quad \text{OR.}$$

$$= \frac{n\Sigma dx\,dy - [\Sigma dx][\Sigma dy]}{\sqrt{n\Sigma (dx)^2 - (\Sigma dx)^2}\sqrt{n\Sigma (dy)^2 - (\Sigma dy)^2}}$$

6.2 Spearman's rank correlation coefficient $r = 1 - \frac{6\Sigma D^2}{n(n^2-1)}$

SOME FORMULAE

Measures	Individual series	Discrete Series	Continuous series	Interpolation formulae
Mean	$\frac{\Sigma x}{n}$	$\frac{\Sigma fx}{N}$ or $a + \frac{\Sigma fd}{N}$	$\frac{\Sigma fx}{N}$ or $a + \frac{\Sigma fd}{N} \times c$	
Median [Q_2]	$\left(\frac{n+1}{2}\right)^{th}$ item if n is odd.	$\left[\frac{N+1}{2}\right]^{th}$ item if N is odd	$\left[\frac{N}{2}\right]^{th}$ item	$l_1 + \frac{\left(\frac{N}{2} - cf\right) \times c}{f}$
Q_1	$\left(\frac{n+1}{4}\right)^{th}$ item	$\left[\frac{N+1}{4}\right]^{th}$ item	$\left[\frac{N}{4}\right]^{th}$ item	$l_1 + \frac{\left[\frac{N}{4} - cf\right] \times c}{f}$
Q_3	$\left[\frac{n+1}{4} \times 3\right]^{th}$ item	$\left[\frac{N+1}{4} \times 3\right]^{th}$ item	$\left[\frac{N}{4} \times 3\right]^{th}$ item	$l_1 + \frac{\left[\frac{3N}{4} - cf\right] \times c}{f}$
P_r	$\left[\frac{n+1}{100} \times r\right]^{th}$ item	$\left[\frac{N+1}{100} \times r\right]^{th}$ item	$\left[\frac{N}{100} \times r\right]^{th}$ item	$l_1 + \frac{\left[\frac{rN}{100} - cf\right] \times c}{f}$
Mode	Item appearing more frequently	Item having highest frequency	Class having highest Frequency	$l_1 + \frac{(f_1 - f_0) \times c}{2f_1 - f_0 - f_2}$

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Measure	Individual series	Discrete series	Continuous series
Standard deviation (SD)	$\sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2}$ or $\sqrt{\frac{\Sigma d^2}{n} - \left(\frac{\Sigma d}{n}\right)^2}$	$\sqrt{\frac{\Sigma f d^2}{N} - \left(\frac{\Sigma f d}{N}\right)^2}$	$\sqrt{\frac{\Sigma f d'^2}{N} - \left(\frac{\Sigma f d'}{N}\right)^2} \times C$
Raw moment (moment about any value M_r')			
M_1'	$\frac{\Sigma d}{n}$	$\frac{\Sigma f d}{N}$	$\frac{\Sigma f d'}{N} \times C$
M_2'	$\frac{\Sigma d^2}{n}$	$\frac{\Sigma f d^2}{N}$	$\frac{\Sigma f d'^2}{N} \times C^2$
M_3'	$\frac{\Sigma d^3}{n}$	$\frac{\Sigma f d^3}{N}$	$\frac{\Sigma f d'^3}{N} \times C^3$
M_4'	$\frac{\Sigma d^4}{n}$	$\frac{\Sigma f d^4}{N}$	$\frac{\Sigma f d'^4}{N} \times C^4$

Range = Highest Value — Lowest Value = H − L

Moment about mean — central moment M_r

$$M_1 = 0$$

$$M_2 = M_2' - (M_1')^2$$

$$M_3 = M_3' - 3M_2'M_1' + 2(M_1')^3$$

$$M_4 = M_4' - 4M_3'M_1' + 6M_2'(M_1')^2 - 3(M_1')^4$$

Karl Pearson's coefficient of skewness = $\frac{\text{Mean} - \text{Mode}}{\text{S.D}}$ OR $\frac{3(\text{Mean} - \text{Median})}{\text{S.D}}$

Bowley's coefficient of skewness = $\frac{Q_3 + Q_1 - 2\text{Median}}{Q_3 - Q_1}$

Moment Measure of skewness = $\frac{M_3}{\sqrt{M_2^3}}$

Moment Measure of Kurtosis = M_4/M_2^2

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Correlation

Introduction

The statistical methods so far studied relate to analysis of one variable only. But we may come across with situations where there is relation between two variables and a statistical analysis is necessary to study such relation. For example, an economist finds that along with the increase in supply of money, there is increase in price level also. In this case he may be interested to study the degree of relation between the two variables namely supply of money and price level. The statistical technique that can be used to study such relationships is the Correlation Analysis.

Definition

Two variables are said to be correlated if the change in one variable results in a corresponding change in the other variable. That is, when two variables move together, we say they are correlated. For example, when the price of a commodity rises, the supply for that commodity also rises. On the other hand, if price falls the supply also falls. So both variables move together or they move in sympathy. Hence price and supply are correlated.

Correlation is defined as "the tendency of two or more groups or series of items to vary together directly or inversely". Boddington states that " Whenever some definite connection exists between the two or more groups, classes or series or data there is said to be correlation ".

Bowley defines correlation as, "when two quantities are so related that the fluctuations in one are in sympathy with the fluctuations of the other, that an increase or decrease of the one is found in connection with an increase or decrease of the other and greater the magnitude of change in one, greater is the magnitude of change in the other, the quantities are held to be correlated".

According to A. M. Tuttle "Correlation is an analysis of the association between two or more variables".

Correlation and Causation

The word correlation usually implies cause and effect relationship, that is, mutual interdependence. For example, a change in the price is the cause for a change in demand. That is, there exists causal connection between the two variables. Causal connection means cause and effect relationship. A positive correlation between the increase in cigarette smoking and the increase in cancer may prove that one causes the other. But correlation does not always imply cause and effect relationship.

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When two variables are correlated there need not be cause and effect relationship. It is just possible that a high degree of correlation between the variables may be due to the same cause affecting each variable. For example, a high degree of correlation between the yield per acre of rice and tea may be due to the fact that both are related to the amount of rainfall. But none of the two variables is the cause of the other.

There may be high degree of correlation between the variables but it may be difficult to pinpoint as to which is the effect. For example, demand and supply, price and production, etc. It is a well known principle of Economics that as the price of a commodity increases its demand goes down and so change in price is the cause and change in demand is the effect. But it is also possible that increased demand of commodity is due to growth of population or other reasons.

Two series may show a high degree of correlation which may result purely from chance also. For example, during the last decade, there has been a significant increase in the sales of news papers and in the number of crimes. That does not mean that there exists any correlation between sales of news paper and crimes. But we can try to establish correlation between the two variables. Such illogical correlation is referred to as nonsensical or spurious correlation. It is a case of casual relationship.

Different kinds of correlation

The different types of correlation are

1. Positive and negative correlation
2. Linear and non - linear correlation
3. Simple, Multiple and Partial correlation

(1) Positive and Negative correlation

Correlation can be either positive or negative. When the values of two variables move in the same direction, correlation is said to be positive ie an increase in the value of one variable results into an increase in the value of the other variable also or if a decrease in the value of one variable, results into a decrease in the value of the other variable also correlation is said to be positive. If on the other hand, the value of two variables move in opposite directions so that an increase in the value of one variable, results in to a decrease in the value of the other variable or the decrease in the value of one variable results into an increase in the value of the other variable the correlation is said to be negative. Generally, price and supply are positively correlated because when price increases supply also increases and when price comes down supply follows. Correlation between price and demand is said to be inverse or negative because with a fall in price, demand goes up and with a rise in price, demand comes down.

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Ex. 1

x	y
30	15
35	20
42	22
60	27
65	30

Ex. 2

x	y
20	30
15	29
10	20
8	15
7	10

Ex. 3

x	y
30	30
35	25
40	20
42	18
45	10

Ex. 4

x	y
30	20
28	25
25	30
18	32
10	40

In example 1, both x and y increase. In example 2, both x and y decrease. In example 3, x increases but y decreases. In example 4, x decreases but y increases. So x and y are positively correlated in examples 1 and 2 while they are negatively correlated in examples 3 and 4.

(2) Linear and Non linear Correlation

Correlation may be linear or non linear. When the amount of change in one variable leads to a constant ratio of change in the other variable, correlation is said to be linear. For example if price goes up by 10% it leads to a rise in supply by 15% each time, then there is a linear relation between price and supply. When there is linear correlation, the point plotted on a graph will give a straight line. Generally we do not find linear correlation in the economic and social phenomena becauase of the influence of multiple and complex factors on the variable.

Ex:

x :	20	25	30	35	40
y :	30	37.5	45	52.5	60

In this example there is linear relationship as there is the ratio 2 : 3 at all points. If we plot these points, they will lie on a straight line.

Correlation is said to be non linear (or curvilinear), when the amount of change in one variable is not in constant ratio to the change in the other variable. In the case of curvilinear correlation, the ratio of change fluctuates and is never constant.

(3) Simple, Partial and Multiple correlation

In the study of relationship between variables, if there are only two variables, the correlation is said to be simple. For example, the correlation between price and demand is simple.

When one variable is related to a number of others, correlation is not simple. Then the study of relationship between the variables is done with the help of partial or multiple correlation.

In the study of multiple correlation we measure the degree of association between one variable on one side and all the other variables together on the other side. Thus the relationship between yield with both rainfall and temperature together is multiple correlation.

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In partial correlation we study the relationship of one variable with one of the other variables presuming that the other variables remain constant. For example, let there be three variables yield, rainfall and temperature. Each is related with the other. Then the relationship between yield and rainfall (assuming temperature is constant) is the partial correlation.

Methods for studying correlation (Measures of Correlation)

Correlation between two variables can be measured by both graphic and algebraic method. Scatter diagram and correlation graph are the two important graphic methods while coefficient of correlation is an algebraic method used for measuring correlation.

(1) Scatter diagram

This is a graphical method of studying correlation between two variables. One of the variables is shown on the X – axis and the other on the Y - axis. Each pair of values is plotted on the graph by means of a dot mark. After, all the items are plotted we get as many dots on the graph paper as the number of points. If these points show some trend either upward or downward, the two variables are said to be correlated. If the plotted points do not show any trend, the two variables are not correlated. If on the other hand the tendency is reverse so that the points show a downward trend from the left top to the right bottom, correlation is negative.

The scatter diagram is a visual aid to show the presence or absence of correlation between two variables. A line of best fit can be drawn using the method of least square. This line will be as close to the points as possible. If the points are falling very close to this line, there is very high degree of correlation . If they lie very much away from this line it shows that the correlation is not much.

For the two sets of data given below draw scatter diagrams and comment on the relationship between variables x and y.

Set I

x	y
5	6
8	10
2	3
4	4
10	12
6	8
8	8

Set II

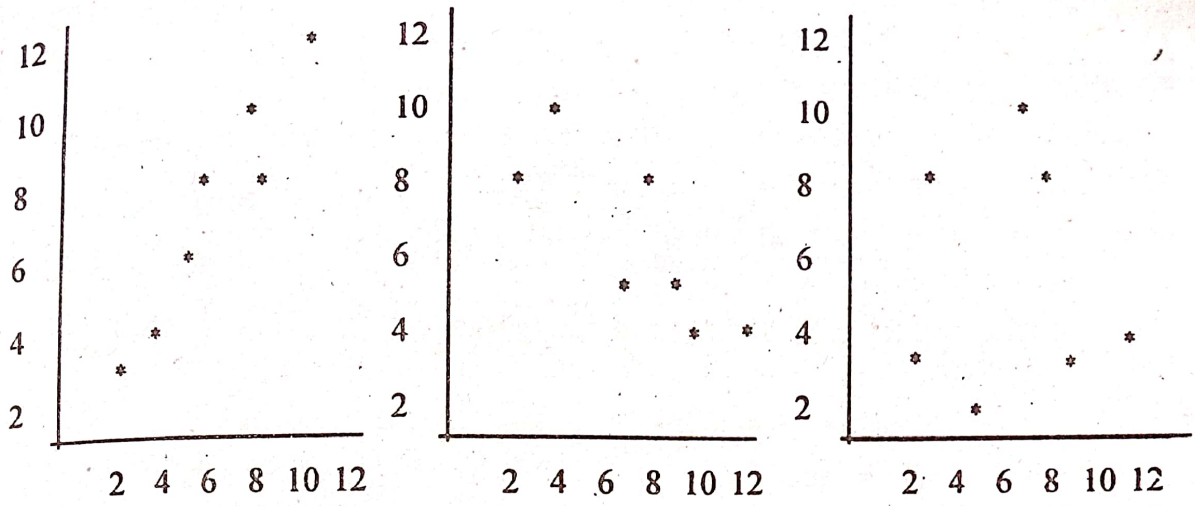
x	y
12	4
7	5
10	4
8	8
4	10
9	5
2	8

Set III

x	y
2	3
5	2
7	10
8	8
3	8
9	3
11	4

Ans: Scatter diagram

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From the three scatter diagrams we can conclude that there is positive correlation between x and y in the I set , there is negative correlation between x and y in the II set and there is no correlation between x and y in the set III.

Merits of Scatter diagram are

1. It is easy to plot the points.
2. It is simple to understand.
3. Abnormal values in the data can be easily detected.
4. The value of dependent variable for a given value of independent variable can be detected.
5. The extreme values do not affect it.

Demerits of Scatter diagram are

1. The degree of correlation cannot be easily estimated.
2. Algebraic treatment is not possible.
3. When the number of pairs of observations is either very big or very small, the method is not easy.

(2) Correlation Graph (Graphic method)

Under this method, separate curves are drawn for the X variable and Y variable on the same graph paper. The values of the variable are taken as ordinates of the poionts plotted. From the direction and closeness of the two curves we can infer whether the variables are related. If both the curves move in the same direction (upward or downward), correlation is said to be positive. If the curves are moving in the opposite direction correlation is said to be negative.

(3) Coefficient of correlation

Coefficient of correlation is an algebraic method of measuring correlation. Under this method, we measure correlation by finding a value known as the coefficient of correlation using an appropriate formula. Correlation coefficient is a numerical value. It shows the degree or the extent of correlation between two variables.

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Coefficient of correlation is a pure number lying between –1 and +1. When the correlation is negative, it lies between –1 and 0. When the correlation is positive, it lies between 0 and 1. When the coefficient of correlation is zero, it indicates that there is no correlation between the variables. When the correlation coefficient is 1, there is perfect correlation. Between no correlation and perfect correlation there are varying degrees of correlation.

Coefficient of correlation can be computed by applying the methods given below:

1. Karl Pearson's method
2. Spearman's method
3. Concurrent deviation method

(1) KARL PEARSON'S COEFFICIENT OF CORRELATION

Of the several methematical methods of measuring correlation the Karl Pearson's method, popularly known as Pearsonian coefficient of correlation, is most widely used in practice. Karl Pearson, the great biologist and statistician has given a formula for calculation of coefficient of correlation. The pearsonian coefficient of correlation is denoted by the symbol 'r'. The formula for computing pearsonial Coefficient of correlation is $r = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{n\,\sigma_x\,\sigma_y}$ where σ_x = Standard deviation of x series. σ_y = Standard deviation of y series and n = Number of pairs of observations. This is also known as product moment correlation coefficient.

The above formula can be expressed in the following form also

$$\frac{n\Sigma xy - (\Sigma x.\Sigma y)}{\sqrt{n\Sigma x^2 - (\Sigma x)^2}\ \sqrt{n\Sigma y^2 - (\Sigma y)^2}}$$

Ex. 1: Calculate coefficient of correlation

x :	2	3	4	5	6	7	8
y :	4	5	6	12	9	5	4

Ans:

x	y	xy	x^2	y^2
2	4	8	4	16
3	5	15	9	25
4	6	24	16	36
5	12	60	25	144
6	9	54	36	81
7	5	35	49	25
8	4	32	64	16
35	45	228	203	343

Coefficient of correlation (r)

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$$= \frac{n\sum xy - (\sum x \cdot \sum y)}{\sqrt{n\sum x^2 - (\sum x)^2}\ \sqrt{n\sum y^2 - (\sum y)^2}}$$

$$= \frac{7 \times 228 - (35 \times 45)}{\sqrt{7 \times 203 - (35)^2}\ \sqrt{7 \times 343 - (45)^2}} = \frac{1596 - 1575}{\sqrt{1421 - 1225}\ \sqrt{2401 - 2025}}$$

$$= \frac{21}{\sqrt{196}\ \sqrt{376}} = \frac{21}{14 \times 19.39} = \frac{21}{271.46} = \underline{0.077}$$

Ex. 2: Calculate Karl Pearson's correlation coefficient between x and y from the following data:

$n = 10$, $\sum x = 35$, $\sum x^2 = 203$, $\sum y = 28$, $\sum y^2 = 140$, $\sum xy = 168$

Ans:

Correlation Coefficient $= \dfrac{n\sum xy - (\sum x \cdot \sum y)}{\sqrt{n\sum x^2 - (\sum x)^2}\ \sqrt{n\sum y^2 - (\sum y)^2}}$

$$= \frac{10 \times 168 - (35 \times 28)}{\sqrt{10 \times 203 - (35)^2}\sqrt{10 \times 140 - (28)^2}} = \frac{192}{\sqrt{805}\sqrt{616}} = \underline{.99}$$

Ex. 3: The data given below pertain to the price and demand for a particular commodity over a period of five years. Compute the correlation coefficient between price and demand.

Price (in Rs/Q) : 7 8 9 6 5

Demand in tons : 8 6 7 9 10

Ans:

x	y	xy	x^2	y^2
7	8	56	49	64
8	6	48	64	36
9	7	63	81	49
6	9	54	36	81
5	10	50	25	100
35	40	271	255	330

Correlation coefficient $= \dfrac{n\sum xy - (\sum x \cdot \sum y)}{\sqrt{n\sum x^2 - (\sum x)^2}\ \sqrt{n\sum y^2 - (\sum y)^2}}$

$$= \frac{5 \times 271 - (35 \times 40)}{\sqrt{5 \times 255 - (35)^2}\ \sqrt{5 \times 330 - (40)^2}} = \frac{1355 - 1400}{\sqrt{1275 - 1225}\ \sqrt{1650 - 1600}}$$

$$= \frac{-45}{\sqrt{50}\ \sqrt{50}} = \frac{-45}{7.07 \times 7.07} = \frac{-45}{50} = \underline{-0.9}$$

Degree of correlation (Interpretation of r)

The degree or the intensity of the relationship between two variables can be ascertained by finding the value of coefficient of correlation. The degree of correlation can be classified into:-

1. **Perfect Correlation:** When the change in the two variables is such that with an increase in the value of one, the value of the other increases in a fixed proportion, correlation is said to be perfect. Perfect correla-

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tion may be positive or negative. Coefficient of correlation is +1 for perfect positive correlation and it is –1 for perfect negative correlation.

2. **No Correlation:** If changes in the value of one variable are not associated with changes in the value of the other variable, there will be no correlation. When there is no correlation the coefficient of correlation is zero.

3. **Limited degree of correlation:** In between perfect correlation and no correlation there may be limited degree of correlation. Limited degree of correlation may also be positive or negative. Limited degree of correlation may be termed as high, moderate or low. For limited degree of correlation the coefficient of correlation lies between 0 and 1 numerically.

Ex. 4: What would be your interpretation if the correlation coefficient 'r' is equal to (1) 0 (2) –1 (3) 1 (4) .2 (5) .9 (6) .52

Ans:

(1) when r = 0, there is no correlation between the variables.
(2) when r = –1, there is negative perfect correlation.
(3) when r = 1, there is perfect positive correlation.
(4) When r = .2, there is very low positive correlation.
(5) When r = .9, there is high positive correlation.
(6) When r = .52, there is moderate positive correlation.

Ex. 5: Compute Karl Pearson's Coefficient of Correlation

Price (Rs) : 11 12 13 14 15 16 17 18 19 20
Demand (Rs) : 30 29 29 25 24 24 24 21 18 15
Also comment on the result obtained.

Price (Rs.) x	Demand (Kg.) y	xy	x^2	y^2
11	30	330	121	900
12	29	348	144	841
13	29	377	169	841
14	25	350	196	625
15	24	360	225	576
16	24	384	256	576
17	24	408	289	576
18	21	378	324	441
19	18	342	361	324
20	15	300	400	225
155	239	3577	2485	5925

$$\text{Coefficient of correlation} = \frac{n\Sigma xy - (\Sigma x \cdot \Sigma y)}{\sqrt{n\Sigma x^2 - (\Sigma x)^2}\ \sqrt{n\Sigma y^2 - (\Sigma y)^2}}$$

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$$= \frac{(10\times3577)-(155\times239)}{\sqrt{10\times2485-(155)^2}\ \sqrt{10\times5925-(239)^2}} = \frac{35770-37045}{\sqrt{24850-24025}\ \sqrt{53250-57121}}$$

$$= \frac{-1275}{\sqrt{825}\ \sqrt{2129}} = \frac{-1275}{28.72\times46.14} = \frac{-1275}{1325.2} = \underline{-.962}$$

There is very high negative correlation between price and demand

The coefficient of correlation is not affected by change of origin. So we can replace x and y in the formula by d_x and d_y. Hence the formula becomes,

$$r = \frac{n\sum dx\,dy - (\sum dx \times \sum dy)}{\sqrt{n\sum dx^2 - (\sum dx)^2}\ \sqrt{n\sum dy^2 - (\sum dy)^2}}$$

where d_x and d_y are deviations taken from assumed averages in the two series.

Ex. 6: Calculate Karl Pearson's coefficient of correlation from the following data using 44 and 26 respectively as the origin of X and Y.

X : 43 44 46 40 44 42 45 42 38 40 42 57

Y : 29 31 19 18 19 27 27 29 41 30 26 10

Ans: Let $d_x = x - 44$ and $d_y = y - 26$

x	y	d_x	d_y	d_xd_y	d_x^2	d_y^2
43	29	–1	3	–3	1	9
44	31	0	5	0	0	25
46	19	2	–7	–14	4	49
40	18	–4	–8	32	16	64
44	19	0	–7	0	0	49
42	27	–2	1	–2	4	1
45	27	1	1	1	1	1
42	29	–2	3	–6	4	9
38	41	–6	15	–90	36	225
40	30	–4	4	–16	16	16
42	26	–2	0	0	4	0
57	10	13	–16	–208	169	256
		–5	–6	–306	255	704

$$\text{Correlation coefficient} = \frac{n\Sigma d_x dy - (\Sigma d_x \Sigma dy)}{\sqrt{n\Sigma d_x^2 - (\Sigma d_x)^2}\ \sqrt{n\Sigma d_y^2 - (\Sigma d_y)^2}}$$

$$= \frac{(12\times-306)-(-5\times-6)}{\sqrt{12\times255-(-5)^2}\ \sqrt{12\times704-(-6)^2}} = \frac{-3672-30}{\sqrt{3060-25}\ \sqrt{8448-36}} = \frac{3702}{\sqrt{3035}\ \sqrt{8412}}$$

$$= \underline{-.73}$$

Ex. 7: In a correlation analysis of 13 pairs of observations of x and y, the following values are obtained. Sum of the deviations of x and y vlues are –117 and –260; sum of the squares of deviations of x

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and y values are 1313 and 6580; sum of the products of deviations of x and y values is 2827. Find coefficient of correlation.

Ans: Given $\Sigma d_x = -117, \Sigma d_y = -260, \Sigma d_x^2 = 1313, \Sigma d_y^2 = 6580, \Sigma d_x d_y = 2827, n = 13.$

$$\text{Correlation Coefficient} = \frac{n\Sigma d_x dy - (\Sigma d_x \Sigma dy)}{\sqrt{n\Sigma d_x^2 - (\Sigma d_x)^2}\ \sqrt{n\Sigma dy^2 - (\Sigma dy)^2}}$$

$$= \frac{13 \times 2827 - (-117 \times -260)}{\sqrt{13 \times 1313 - (-117)^2}\ \sqrt{13 \times 6580 - (-260)^2}} = .813$$

If we can take deviation from actual mean then, $d_x = x - \bar{x}$ and $dy = y - \bar{y}$ so that $\Sigma d_x = 0, \Sigma d_y = 0$. Then the formula becomes $\dfrac{\Sigma d_x d_y}{\sqrt{\Sigma d_x^2}\ \sqrt{\Sigma d_y^2}}$

Ex. 8: Find the coefficient of correlation from the following data

x	:	12	20	15	22	18	24	20	12	15	22
y	:	30	35	28	36	29	39	30	25	30	38

Ans: $\bar{x} = \dfrac{\Sigma x}{n} = \dfrac{180}{10} = 18$ and $\bar{y} = \dfrac{\Sigma y}{n} = \dfrac{320}{10} = 32$

Taking deviation from $\bar{x}$ and $\bar{y}$ ie $d_x = x - 18$ and $d_y = y - 32$

x	y	d_x	d_y	$d_x d_y$	d_x^2	d_y^2
12	30	–6	–2	12	36	4
20	35	+2	+3	6	4	9
15	28	–3	–4	12	9	16
22	36	+4	+4	16	16	16
18	29	0	–3	0	0	9
24	39	+6	+7	42	36	49
20	30	+2	–2	–4	4	4
12	25	–6	–7	42	36	49
15	30	–3	–2	6	9	4
22	38	+4	+6	24	16	36
180	320	0	0	156	166	196

$$r = \frac{\Sigma d_x d_y}{\sqrt{\Sigma d_x^2}\ \sqrt{\Sigma d_y^2}} = \frac{156}{\sqrt{166}\ \sqrt{196}} = .87$$

Ex. 9: Sum of the product of the deviations of the variables x and y from their respective means is 1320, sum of the squares of deviations of x and y variables from their respective means are 628 and 3720. Find the coefficient of correlation.

Ans: $\Sigma d_x d_y = 1320, \Sigma d_x^2 = 628, \Sigma d_y^2 = 3720$

When deviations are taken from actual mean Σd_x and Σd_y will be zero so that the formula becomes

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$$r = \frac{\Sigma d_x d_y}{\sqrt{\Sigma d_x^2}\ \sqrt{\Sigma d_y^2}} = \frac{1320}{\sqrt{628}\ \sqrt{3720}} = \underline{\underline{0.86}}$$

Note: Coefficient of correlation is not affected by change of scale. Therefore if we multiply or divide all the values of a variable by a constant, the coefficient of correlation will not change.

Ex. 10: Find the coefficient of correlation between x and y and interpret the result.

X :	1.2	1.1	1.9	1.8	1.0	0.9
Y :	15	10	20	10	10	5

Ans: Multiplying x values by 10 and divide y values by 5. Then we get new x values and y values.

x	y	xy	x^2	y^2
12	3	36	144	9
11	2	22	121	4
19	4	76	361	16
18	2	36	324	4
10	2	20	100	4
9	1	9	81	1
79	14	199	1131	38

$$\text{Coefficient of correlation} = \frac{n\Sigma xy - (\Sigma x \cdot \Sigma y)}{\sqrt{n\Sigma x^2 - (\Sigma x)^2}\ \sqrt{n\Sigma y^2 - (\Sigma y)^2}}$$

$$= \frac{6\times199 - (79\times14)}{\sqrt{6\times1131 - (79)^2}\ \sqrt{6\times38 - (14)^2}} = \frac{1194-1106}{\sqrt{6786-6241}\ \sqrt{228-196}} = \frac{88}{\sqrt{545}\ \sqrt{32}} = \underline{\underline{0.67}}$$

$\therefore$ There is good positive correlation between x and y.

Ex. 11: Find coefficient of correlation between age and playing habit of the following students.

Age :	14.5-15.5	15.5-16.5	16.5-17.5	17.5-18.5	18.5-19.5	19.5-20.5
No. of students:	250	200	150	120	100	80
Regular Players:	200	150	90	48	30	12

Ans: Hint: 1) Playing habit in % $= \frac{\text{No. of regular players}}{\text{No. of students}} \times 100$

2) Take the middle value for age.

Age (x)	Playing habit % (y)	d_x (x – 18)	d_y (y – 50)	$d_x d_y$	d_x^2	d_y^2
15	80	–3	30	–90	9	900
16	75	–2	25	–50	4	625
17	60	–1	10	–10	1	100
18	40	0	–10	0	0	100
19	30	1	–20	20	1	400
20	15	2	–35	4	4	1225
		–3	0	–240	19	3350

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$$r = \frac{n\Sigma d_x d_y - (\Sigma d_x \Sigma d_y)}{\sqrt{n\Sigma d_x^2 - (\Sigma d_x)^2}\ \sqrt{n\Sigma d_y^2 - (\Sigma d_y)^2}}$$

$$= \frac{6\times -240 - (-3\times 0)}{\sqrt{6\times 19 - (-3)^2}\ \sqrt{6\times 3350 - (0)^2}} = \frac{-1440 - 0}{\sqrt{114 - 9}\ \sqrt{20100}}$$

$$= \frac{-1440}{10.25\times 141.77} = \frac{-1440}{1453.14} = \underline{\underline{-.99}}$$

Ex. 12: With the following data in 6 cities calculate the coefficient of correlation by Pearson's method between the density of population and the death rate.

	Area in sq. miles	Population in '000	No. of deaths
A	150	30	300
B	180	90	1440
C	100	40	560
D	60	42	840
E	120	72	1224
F	80	24	312

Ans:

Desnsity = $\frac{\text{Population}}{\text{Area}}$ and Death rate = $\frac{\text{No. of deaths}}{\text{Population}}$

Density x	Death rate y	d_x $(x-4)\times 10$	d_y $(y-16)$	$d_x d_y$	d_x^2	d_y^2
.2	10	–2	–6	12	4	36
.5	16	1	0	0	1	0
.4	14	0	–2	0	0	4
.7	20	3	4	12	9	16
.6	17	2	1	2	4	1
.3	13	–1	–3	3	1	9
		3	–6	29	19	66

Correlation coefficient between density of population and the death rate

$$= \frac{n\Sigma d_x d_y - (\Sigma d_x \Sigma d_y)}{\sqrt{n\Sigma d_x^2 - (\Sigma d_x)^2}\ \sqrt{n\Sigma d_y^2 - (\Sigma d_y)^2}}$$

$$= \frac{6\times 29 - (3\times -6)}{\sqrt{6\times 19 - (3)^2}\ \sqrt{6\times 66 - (-6)^2}} = \frac{192}{\sqrt{105}\ \sqrt{360}} = \underline{\underline{.98}}$$

Properties of Correlation Coefficient

1. Correlation coefficient has a well defined formula.
2. Correlation coefficient is a pure number and is independent of the units of measurement.
3. It lies between –1 and +1.
4. Correlation coefficient does not change with reference to change of origin or change of scale.

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5. Coefficient of correlation between x and y is same as that between y and x.

Probable Error

Probable error of the coefficient of correlation is a statistical measure which measures reliability and dependability of the value of coefficient of correlation. If probable error is added to or subtracted from the coefficient of correlation it would give two such limits within which we can reasonably expect the value of coefficient of correlation to vary. Usually the coefficient of correlation is calculated from samples. For different samples drawn from the same population the coefficient of correlation may vary. But the numerical value of such variations is expected to be less than the probable error.

The formula for finding probable error is

$$\text{Probable Error} = \frac{.6745\,(1-r^2)}{\sqrt{n}}$$ where 'r' stands for the correlationcoefficient and 'n' number of pairs of observation.

The quantity $\frac{1-r^2}{\sqrt{n}}$ is known as standard error of correlation coefficient.

Ex. 13: If r = .6 and n = 64, find Probable Error and Standard error.

Ans: $r = .6$

$$\text{P. E.} = \frac{.6745\,(1-r^2)}{\sqrt{n}} = \frac{.6745\,(1-.36)}{\sqrt{64}} = \frac{.6745 \times .64}{8} = \underline{\underline{.054}}$$

$$\text{Standard error} = \frac{1-r^2}{\sqrt{n}} = \frac{1-.36}{\sqrt{64}} = \underline{\underline{.08}}$$

Interpretation of coefficient of correlation on the basis of probable error.

(1) If the coefficient of correlation is less than its probable error, it is not at all significant.

(2) If the coefficient of correlation is more than six times its probable error, it is significant.

(3) If the probable error is not much and if the coefficient of correlation is .5 or more it is generally considered to be significnat.

Ex. 14: The following table gives the result of the S.S.L.C Examination held in 1974.

Age of candidates :	13	14	15	16	17	18	19	20	21
Percentage of failures:	39	41	43	34	37	39	49	47	55

Calculate coefficient of correlation and estimate probable error. From the result can you definitely assert that failure is correlated with age ?

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Ans: Taking $x - 17 = d_x$ and $y - 43 = d_y$

Age (x)	Failures (y)	d_x	d_y	$d_x d_y$	d_x^2	d_y^2
13	39	–4	–4	16	16	16
14	41	–3	–2	6	9	4
15	43	–2	0	0	4	0
16	34	–1	–9	9	1	81
17	37	0	–6	0	0	36
18	39	1	–4	–4	1	16
19	49	2	6	12	4	36
20	47	3	4	12	9	16
21	55	4	12	48	16	144
		0	–3	99	60	349

$$\text{Coefficient of correlation} = r = \frac{n\Sigma d_x d_y - (\Sigma d_x \Sigma d_y)}{\sqrt{n\Sigma d_x^2 - (\Sigma d_x)^2}\ \sqrt{n\Sigma d_y^2 - (\Sigma d_y)^2}}$$

$$= \frac{9 \times 99 - (0 \times -3)}{\sqrt{9 \times 60 - (0)^2}\ \sqrt{9 \times 349 - (-3)^2}} = \frac{891}{\sqrt{540}\ \sqrt{3132}} = \underline{.6852}$$

$$\text{P. E} = \frac{.6745\ (1-r^2)}{\sqrt{n}} = \frac{.6745\ (1-.4695)}{\sqrt{9}} = \frac{.6745 \times .5305}{3} = \underline{.1193}$$

Since the probable error is less and correlation is more than 6 times the probable error, correlation is significant. Therefore we can assert that failure is correlated with age.

Ex. 15: Coefficient of correlation between two variables is calculated to be –.98. Find the value of probable error and hence interpret the result (n = 10). Find the limits within which population correlation coefficient may lie:

Ans:

$$\text{Probable error} = \frac{.6745(1-r^2)}{\sqrt{n}} = .6745\ \frac{1-(-.98)^2}{\sqrt{10}} = \frac{.6745 \times .0396}{3.162}$$

$$= .008447$$

Probable error is very small and the correlation coefficient is very high negative. Therefore there is significant negative correlation.

Population correlation coefficient $= r \pm PE = -.98 \pm .008$

$= -.98 - .008, -.98 + .008 = \underline{\underline{(-.988, -.972)}}$

Ex. 16: If r = .89, P. E = .023 find the value of n

Ans:

$$\text{P.E} = \frac{.6745(1-r^2)}{\sqrt{n}} \qquad \therefore \sqrt{n} \times \text{P.E} = .6745(1-r^2)$$

$$\therefore \sqrt{n} = \frac{.6745(1-r^2)}{\sqrt{n}} = \frac{.6745(1-.7921)}{.023} = \frac{.6745 \times .2079}{.023} = 6$$

$$\therefore n = 6^2 = \underline{36}$$

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(2) RANK CORRELATION

Karl Pearson's Coefficient is applicable when variables are measured in quantitative form. But in many cases measurement is not possible because they are in quantitative form. For example, we cannot measure the beauty or intelligence quantitatively. But it may be possible, in their case, to rank the individuals in some order. The correlation coefficient obtained from the ranks so obtained is called rank correlation.

Therefore rank correlation is the correlation obtained from ranks, instead of their quantitative measurement.

Thus, when the values of two variables are expressed in ranks and therefrom correlation is obtained, that correlation is known as rank correlation. Spearman has devised a formula known as Spearman's rank correlation coefficient to find the correlation coefficient from the ranks. According to spearman's method, the formula for Rank Correlation Coefficient is $1 - \frac{6 \Sigma D^2}{n(n^2-1)}$

where 'D' is the difference between ranks and n, number of items.

Ex.17: The ranking of 10 individuals at the start and at the finish of a course of a training are as follows

Individuals :	**A**	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**	**J**
Rank before :	**1**	**6**	**3**	**9**	**5**	**2**	**7**	**10**	**8**	**4**
Rank after :	**6**	**8**	**3**	**2**	**7**	**10**	**5**	**9**	**4**	**1**

Calculate Spearmen's Rank correlation coefficient.

Ans:

Rank before	Rank after	Rank difference (D)	D^2
1	6	5	25
6	8	2	4
3	3	0	0
9	2	7	49
5	7	2	4
2	10	8	64
7	5	2	4
10	9	1	1
8	4	4	16
4	1	3	9
			176

Rank correlation coefficient $= 1 - \frac{6 \Sigma D^2}{n(n^2-1)} = 1 - \frac{6 \times 176}{10(100-1)}$

$= 1 - \frac{1056}{990} = 1 - 1.07 = \underline{\underline{-0.07}}$

Ex. 18: Ten competitors in a beauty contest are ranked by three judges in the following order.

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First Judge :	1	6	5	10	3	2	4	9	7	8
Second Judge:	3	5	8	4	7	10	2	1	6	9
Third Judge :	6	4	9	8	1	2	3	10	5	7

Use the correlation coefficient to discuss which pair of judges have nearest approach to common tastes in beauty.

Ans:

First judge	Second judge	Rank difference (D)	D^2
1	3	2	4
6	5	1	1
5	8	3	9
10	4	6	36
3	7	4	16
2	10	8	64
4	2	2	4
9	1	8	64
7	6	1	1
8	9	1	1
			200

$$\text{Rank Correlation Coefficient} = 1 - \frac{6\sum D^2}{n(n^2-1)} = 1 - \frac{6\times 200}{10(100-1)} = 1 - 1.21 = -.21$$

First judge	Third judge	Rank difference D	D^2
1	6	5	25
6	4	2	4
5	9	4	16
10	8	2	4
3	1	2	4
2	2	0	0
4	3	1	1
9	10	1	1
7	5	2	4
8	7	1	1
			60

$$\text{Rank Correlation Coefficient} = 1 - \frac{6\sum D^2}{n(n^2-1)} = 1 - \frac{6\times 60}{10(100-1)} = 1 - .364 = .636$$

Second judge	Third judge	Rank difference (D)	D^2
3	6	3	9
5	4	1	1
8	9	1	1
4	8	4	16
7	1	6	36
10	2	8	64
2	3	1	1
1	10	9	81
6	5	1	1
9	7	2	4
			214

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Rank Correlation coefficient $= 1 - \frac{6\Sigma D^2}{n(n^2-1)} = 1 - \frac{6\times214}{990} = 1 - 1.30$
$= \underline{-0.30}$

The rank correlation coefficient in the case of I and III Judges is greater than the other two pairs. Therefore I and III judges have highest similarity of thought and have the nearest approach to common taste in beauty.

Ex. 19: Find the rank correlation coefficient between poverty and overcrowding from the table below:

Town	:	A	B	C	D	E	F	G	H	I	J
Poverty	:	17	13	15	16	6	11	14	9	7	12
Over crowding	:	36	46	35	24	12	18	27	22	2	8

Ans:

Town	Poverty		Overcrowding		Difference	D^2
	Values	Rank	Values	Rank	D	
A	17	1	36	2	1	1
B	13	5	46	1	4	16
C	15	3	35	3	0	0
D	16	2	24	5	3	9
E	6	10	12	8	2	4
F	11	7	18	7	0	0
G	14	4	27	4	0	0
H	9	8	22	6	2	4
I	7	9	2	10	1	1
J	12	6	8	9	3	9
						44

Rank correlation coefficient =

$$1 - \frac{6\Sigma D^2}{n(n^2-1)} = 1 - \frac{6\times44}{10(100-1)} = 1 - \frac{264}{990} = 1 - .267 = \underline{0.733}$$

Repeated Rank (Tie in Rank)

When the values repeat in one or both the series, Rank correlation coefficient is obtained using the formula, $1 - \frac{6(\Sigma D^2 + \frac{m^3-m}{12})}{n(n^2-1)}$

where 'm' stands for number of times each value repeats.

Ex. 20: Obtain the rank correlation coefficient for the following data

X :	68	64	75	50	64	80	75	40	55	64
Y :	62	58	68	45	81	60	68	48	50	70

Ans:

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X	Rank of x	y	Rank of y	Rank difference, (D)	D^2
68	4	62	5	1	1
64	6	58	7	1	1
75	2.5	68	3.5	1	1
50	9	45	10	1	1
64	6	81	1	5	25
80	1	60	6	5	25
75	2.5	68	3.5	1	1
40	10	48	9	1	1
55	8	50	8	0	0
64	6	70	2	4	16
					72

$\therefore \Sigma D^2 = 72$

75 Occurs 2 times $\therefore$ $m = 2$ $\quad m^3 - m = 2^3 - 2 = 6$

64 Occurs 3 times $\therefore$ $m = 3$ $\quad m^3 - m = 3^3 - 3 = 24$

68 Occurs 2 times $\therefore$ $m = 2$ $\quad m^3 - m = 2^3 - 2 = 6$

$\therefore$ Total $m^3 - m = 6 + 24 + 6 = 36$

$\therefore$ Rank correlation coefficient

$$= 1 - \frac{6(\Sigma D^2 + \frac{m^3 - m}{12})}{n(n^2 - 1)} = 1 - \frac{6 \times (72 + \frac{36}{12})}{10(100 - 1)} = 1 - \frac{6 \times 75}{990} = .545$$

How to assign rank when there is repetition ?

Ans: Give the average of ranks. For example: In the last example, highest value in the x - series is 80. So it is given rank I. Next two values are the same 75. They are given the average of 2 and 3 ie 2.5. Then 4th rank is given to the next highest value 68. Then 64 occurs 3 times. So each 64 is given the average of ranks 5, 6, 7 ie 6 and so on.

Ex. 21: The ranks of 11 students in two sets are given below. Calculate the coefficient of correlation by the method of rank difference.

Test I : 80 45 55 58 55 60 45 68 70 45 85

Test II : 82 56 50 43 56 62 64 65 70 64 90

Ans:

Ranks			
Test I	Test II	D	D^2
2	2	0	0.00
10	8.5	1.5	2.25
7.5	10	2.5	6.25
6	11	5.0	25.00
7.5	8.5	1.0	1.00
5	7	2.0	4.00
10	5.5	4.5	20.25
4	4	0.0	0.00
3	3	0.0	0.00
10	5.5	4.5	20.25
1	1	0.0	0.00
			79

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55 repeats 2 times $\therefore$ $m = 2$ $\quad\therefore m^3 - m = 6$

45 repeats 3 times $\therefore$ $m = 3$ $\quad\therefore m^3 - m = 24$

64 repeats 2 times $\therefore$ $m = 2$ $\quad\therefore m^3 - m = 6$

56 repeats 2 times $\therefore$ $m = 2$ $\quad\therefore m^3 - m = 6$

Total $m^3 - m = 6 + 24 + 6 + 6 = 42$

$$r = 1 - \frac{6(\Sigma D^2 + \frac{m^3 - m}{12})}{n(n^2-1)} = 1 - \frac{6\times(79 + \frac{42}{12})}{11(121-1)} = 1 - \frac{6\times(79+3.5)}{11\times120}$$

$$= 1 - \frac{6\times82.5}{11\times120} = 1 - 0.375 = \underline{\underline{.625}}$$

Ex. 22. The coefficient of rank correlation of the marks obtained by 10 students in Statistics and Accountancy was 0.2. It was later discovered that the difference in ranks in the two subjects of one of the students was wrongly taken as 7 instead of 9. Find the correct result.

Ans: $r = 0.2$

$$\therefore 1 - \frac{6\Sigma D^2}{n(n^2-1)} = 0.2 \qquad \therefore 1 - .02 = \frac{6\Sigma D^2}{n(n^2-1)}$$

$$\frac{6\Sigma D^2}{10(100-1)} = 0.8 \qquad \therefore \frac{6\Sigma D^2}{990} = 0.8$$

$$\therefore 6\Sigma D^2 = .8 \times 990 = 792$$

$$\Sigma D^2 = 792 \div 6 = 132$$

Correct $\Sigma D^2 = = 132 - 7^2 + 9^2 = 164$

$$\therefore \text{Correct rank correlation coefficient} = 1 - \frac{6\Sigma D^2}{n(n^2-1)} = 1 - \frac{6\times164}{10(100-1)}$$

$$= 1 - .99 = \underline{\underline{0.01}}$$

Merits and Demerits of Rank correlation

Merits

1. It is easy to calculate.
2. It is simple to understand.
3. It can be applied to both quantitative and qualitative data.

Demerits

1. Rank correlation coefficient is only approximate measure as the actual values are not used.
2. It is not convenient when 'n' is large.
3. Further algebraic treatment is not possible.

CONCURRENT DEVIATION METHOD

This is a very simple method of measuring correlation. Under this method the directions of deviations are only taken. The magnitudes of the values are ignored. So this method is useful when we are interested in studying correlation between two variables in a casual manner and are

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